C31 Organic Synthesis – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Reagents and Conditions 1 **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Alkene → alcohol (industrial)				Steam, H_3PO_4, 300 °C, 6 MPa
2	Alcohol → alkene				Conc H_2SO_4 or H_3PO_4, heat
3	Halogenoalkane → alcohol				NaOH(aq), warm
4	Halogenoalkane → nitrile				KCN in ethanol/water, reflux
5	Nitrile → amine				$LiAlH_4$ or H_2/Ni
6	Primary alcohol → aldehyde				Acidified $K_2Cr_2O_7$, distil
7	Primary alcohol → carboxylic acid				Acidified $K_2Cr_2O_7$, reflux
8	Aldehyde or ketone → alcohol				$NaBH_4$
9	Alkene → bromoalkane				HBr
10	Benzene → nitrobenzene				Conc HNO_3 + conc H_2SO_4, below 55 °C

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C31 Organic Synthesis – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 2: Reagents and Conditions 2 **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Nitrobenzene → phenylamine				Sn + conc HCl
2	Benzene → phenyl ketone				Acyl chloride + $AlCl_3$
3	Alkane → chloroalkane				Cl_2, UV light
4	Halogenoalkane → alkene				KOH in ethanol, hot
5	Halogenoalkane → amine				Excess NH_3 in ethanol, heat, sealed tube
6	Aldehyde or ketone → hydroxynitrile				KCN, then dilute acid
7	Carboxylic acid + alcohol → ester				Conc H_2SO_4 catalyst, warm
8	Acyl chloride → ester				Alcohol, room temperature
9	Acyl chloride → amide				NH_3 or amine
10	Alkene → dibromoalkane				Br_2

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.